

In-Situ SOP Materials Inventory

This inventory covers 24 of the 25 SOPs currently listed in In_Situ_SOPs/README.md. Load and Run the Oven Tray is listed in that README but is not yet itemised here. Quantities are per episode unless noted otherwise. Fixed installations are separated from movable tools and consumable or input materials so the list can be used for procurement and station setup.

Materials by task

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Feed a Conveyor Line Station	Stub conveyor with running belt; evenly spaced white pitch marks; near and far rails; overhead light bar and conduit; entry and transfer flags; takt lamp; read-only count panel; fixed load shelf; orientation nest with rib	Parts tote; reject bin	10 square-based housing parts , plus 1 misfeed part pre-positioned on the belt (11 parts total if the misfeed is additional to the tote quantity)
Fit a Fitted Sheet to the Mattress	Bed frame with a cot-size mattress lying square and firm on it; no rail, headboard, or footboard rising above the mattress top; clear access to the top face, all four side faces, and the bottom edge; clear floor along the near long side for the base	None beyond the installed bed and mattress	1 cot-size fitted sheet cut to fit the mattress, with unbroken elastic hem all round, staged spread on the mattress pockets down and unhooked
Grease a Machine Round	Stopped and isolated machine; lockout tag; 3 same-size grease zerks; guard shelf; frame rail; hose clip; tool shelf; fixed grease-gun cradle; card holder	Primed lever grease gun with hose and fitting coupler; blue wipe rag; red purge rag; grease pencil	Grease sufficient for 9 full strokes (3 strokes at each of 3 zerks); lube card with at least 1 empty three-box row
Load the Dishwasher	Installed dishwasher with working latch, open detergent dispenser, control panel, upper and lower racks, and clipped cutlery basket; draining board; utensil cup	None beyond the installed racks, basket, draining board, and utensil cup	2 plates, 2 bowls, 2 cups, 2 forks, 2 knives, 2 spoons; 1 dishwasher detergent pod

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Mop a Floor Zone	Defined hard-floor bay with back kick rail and left/right plinth ends; fixed mop rest	Rectangular bucket with wringer and fill line; handled water jug; flat mop with fitted absorbent pad; folding A-frame caution sign	Clean water: 1 jug , measured to the bucket fill line; 1 clean, dry mop pad fitted to the mop
Refill the Kettle and Dispenser at the Sink	Sink with working cold-water mixer tap; counter; kettle power base; fixed dispenser with open-top filler tank, front fill line, own tap, and drip tray	Electric kettle; separate fill jug with handle, pouring lip, and fill line; absorbent cloth	Cold tap water sufficient to fill the partly filled kettle to its line and to fill the jug once so the partly filled dispenser tank reaches its line
Organize the Freezer In Situ	Installed freezer with open left-hinged door on a working door catch; working shelf with a shelf directly above; back wall; clear floor space in front for the work stand	Work stand; flat work tray with a bag spot and a box spot; 2 open-topped bins with front rails, one labeled BAGS in blue and one BOXES in green	6 dated items: 3 sealed freezer bags (dates 1, 2, 3) and 3 rigid boxes (dates 1, 2, 3), each with a readable date tag and each bag with a blank label patch; 1 label sheet carrying exactly 2 labels , naming bag 2 and bag 3
Restock the Pantry Shelves	Pantry shelf unit with working shelf and shelf directly above; three zone labels (CARTONS, JARS, CANS) fixed to the shelf front edge; clear floor space in front for the stand	Stand; plain cardboard delivery box with no top flaps and untaped, untucked bottom flaps (collapsible)	Older stock already in place: 1 carton, 1 jar, 1 can. New stock: 2 cartons, 2 jars, 2 cans (6 new items); 1 delivery box flattened per episode
Restock the Refrigerator In Situ	Installed refrigerator with open left-hinged door; working shelf; crisper drawer; tall and short door bins; clear adjacent counter	Stock tote	Older stock already in place: 1 carton, 1 jar, 1 tub, 1 produce pack, 1 bottle, 1 small jar. New stock: 1 carton, 1 jar, 1 tub, 2 produce packs, 1 bottle, 1 small jar (7 new items)

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Run a Microwave Heating Cycle	Installed microwave with turntable, front controls, working door/latch, and clear heat-proof counter	Heat-proof plate with broad rim; vented dome cover with tab; food tub; spoon	Thick prepared food sufficient for 3 level spoonfuls , with some food remaining in the tub
Squeegee a Mirror Panel	Fixed framed mirror panel; clear counter with bottle, cloth, and squeegee positions	Trigger spray bottle set to fan spray; straight rubber-blade squeegee; absorbent lint-free cloth	Glass cleaner sufficient for at least 6 trigger bursts
Stage Outbound Lanes	Fixed dock deck with a smooth top and an empty upper shelf directly above its lane strip; 2 route lanes taped as 3 numbered spots each (Lane A from the left end, Lane B from the right end); ROUTE A and ROUTE B signs on the front edge of the upper shelf; 2 hinged lane flags with a catch (OPEN flat / STAGED standing) at the head end of each lane; fixed manifest card holder at the middle of the upper shelf edge	None beyond the installed lanes, flags, and holder	6 same-size closed cases (shoebox size) with front-face route labels A1, A2, A3, B1, B2, B3, reused; A3 and B1 weighted and carrying a red HEAVY band; 1 manifest card (ROUTE A: 3 cases, ROUTE B: 3 cases, heavy first, then by stop), reused
Tend a 3D Printer	Installed 3D printer with magnetic bed, locating pins, parked toolhead, and working control panel; plate rest/work area	Flexible steel build plate; debris tray; stiff plastic scraper; soft brush	1 completed printed part attached to the plate, with skirt and purge line; 1 next print job ready in the printer queue
Tidy a Bookshelf	Bookcase with working shelf, shelf above, fixed right end panel, and lower shelf	Open bin; damp cleaning cloth; solid freestanding bookend	6 books , all visibly different heights; 3 non-book objects : 1 mug, 1 small picture frame, and 1 coiled cable; visible removable dust on the shelf lip

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Unload the Dishwasher to the Cabinet	Installed dishwasher with working upper and lower racks and clipped cutlery basket, door open flat; open-fronted two-shelf cabinet standing on the counter immediately right of the machine, with 2 up-pointing mug hooks fixed under the front edge of the upper shelf; counter drawer directly below the cabinet, drawn open past the counter edge	Three-compartment cutlery drawer insert (forks, knives, spoons)	2 plates, 2 bowls, 2 mugs with handles, 2 forks, 2 knives, 2 spoons , all clean and dry, staged in the racks; no consumables
Unload the Dryer	Installed front-loading dryer on a pedestal or stack with the drum mouth about level with the arms, left-hinged door that stays open on its own and latches when pressed, pull-up lint filter with a tab in a slot at the bottom of the door opening; low table in front of the dryer below the door opening; hamper stand to the right of the drum mouth, level with the table and clear of the door swing	Small light open-top hamper with one stiff hoop carry handle across the top; open lint bin	6 garments: 3 cotton t-shirts and 3 hand towels , dry, loose on the drum floor; 1 lint sheet per episode pressed onto the filter mesh
Handheld Vacuum Pass and Dock	Fixed raised platform in front of the base at about work-table height; small flat short-pile rug with a bound edge and non-slip backing lying on it, left binding at the middle of the base; fixed dock cradle with a spring clip at the platform's right end, holding the vacuum level	Cordless handheld vacuum with a pistol handle, latching power button and power light on top, clear dust cup with an underside flap and a left-side release catch, fixed wide flat nozzle; open-top bin on the platform's left	1 light, even scatter of dry crumbs over the whole rug per episode; vacuum charged enough for three lanes and one edge pass at full power

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
5S Reset a Workbench Zone	Fixed workbench with a wall shelf directly above its back strip; bench power strip with an ON/OFF switch and light; zone tag; tool trough fixed along the back strip with 3 labelled open-front slots (driver, pliers, wrench); consumable rack fixed at the right end of the back strip with 2 labelled open-front bays (glove, wipe); checklist bracket bolted at the right end of the front strip; marker clip; 2 marked cloth home spots	Open-top returns bin; screwdriver, pliers, and wrench; 2 damp wipe cloths; marker	3 strays (1 mug, 1 stapler, 1 small parts box), reused; 2 consumable packs (1 glove box, 1 wipe pack), reused; 1 checklist card with at least 1 empty four-box row; a light scatter of loose dust over the working area per episode
Dress the Bed Pillows and Throw	Made bed with flat tucked sheet and blanket; fixed headboard at the head end; clear floor along the near long side for the base	None beyond the installed bed	2 bare pillows (same size, stacked at the config's pillow spot), 2 pillowcases (same size and colour, folded at the middle of the bed), 1 throw longer than the bed is wide (folded in half at the foot end); all reused
Extract a Part from an Injection Mold Mock	Two half mold mock bolted to a bench, moving half on two guide bars with a front handle; rubber tool mat centred in front of the mold	Flush cutters; pocketed parts tray; scrap cup; reject bin (cutters and output group placed per config)	1 molded plastic part with runner and gate loaded in the cavity per episode; scrap runners for setup test cuts
Insert a Duvet into Its Cover	Bed frame with headboard and mattress wearing a tucked fitted sheet; clear floor along the near long side	None beyond the installed bed	1 duvet (folded in half at the config's duvet spot) and 1 buttoned duvet cover that fits it (hand-gathered into a band at the head end); reused

Task	Fixed equipment / station	Reusable tools and containers	Consumables / input materials
Operate a Press Brake Mock	Hand lever press brake mock bolted to a bench, lever on the right, die groove, two back stops, set to 90 degrees; blank tray left, finished tray and reject bin right	90 degree angle gauge; rubber gauge mat (placed per config); pocketed finished tray; reject bin	1 flat sheet blank per episode with three numbered bend lines, stacked in the blank tray
Reset the Entryway	Bench against a wall under a fixed hook board with 2 coat hooks and 1 bag hook; shallow bowl fixed at the back right; low one shelf shoe rack on the floor at the left end; floor mat between the base and the bench	None beyond the bowl, the rack, and the mat	2 jackets with collar loops, 1 pair of shoes, 1 key set on a ring, 1 bag with a strap ; all reused, staged per config
Seasonal Wardrobe Swap	Bench in front of a wardrobe; fixed short hanging rail above the left end; open shelf above the right end; empty stow bay under the shelf	Lidded storage box with a low divider; 2 hangers	4 soft out-of-season items (stacked per config) and 4 in-season items (2 on hangers, 2 folded flat, in the far half of the box); all reused

Consolidated preparation list

Installed stations

- 1 workbench zone station: fixed bench with a wall shelf above the back strip, power strip with switch and light, zone tag, 3-slot tool trough, 2-bay consumable rack, checklist bracket, marker clip, and 2 marked cloth home spots
- 1 conveyor station
- 1 dock deck station: smooth-top deck with an empty upper shelf over the lane strip, 2 taped route lanes of 3 numbered spots, 2 route signs, 2 hinged lane flags with catches, and a fixed manifest card holder
- 1 bed station with a cot-size mattress on its frame
- 1 isolated machine with 3 grease zerks
- 1 dishwasher station
- 1 defined hard-floor mopping bay
- 1 sink, kettle, and dispenser station
- 1 pantry shelf-unit station with a shelf above and three zone labels

- 1 refrigerator station
- 1 installed freezer station with a two-level shelf
- 1 microwave station
- 1 fixed mirror and counter station
- 1 3D-printer station
- 1 two-level bookshelf station
- 1 dishwasher unload station: dishwasher, countertop two-shelf cabinet with 2 mug hooks, and cutlery drawer
- 1 dryer unload station: front-loading dryer at arm height, low table in front of it, and hamper stand to its right
- 1 bed station with headboard, made bed, pillows, cases, and a throw (pillows and throw)
- 1 injection mold mock bolted to a bench, with centred tool mat
- 1 bed station with headboard and fitted-sheet mattress (duvet)
- 1 press brake mock bolted to a bench, lever on the right, blank tray left, finished tray right
- 1 entryway bench with hook board (3 hooks), bowl, floor shoe rack, and floor mat
- 1 wardrobe bench with rail above the left end, shelf above the right end, and stow bay
- 1 handheld vacuum station: raised platform with a small bound-edge rug, fixed dock cradle with a spring clip at the right end, and an open-top bin at the left

Reusable tools and staging items

- Outbound lanes: none beyond the installed lanes, flags, and manifest holder
- Workbench 5S: open-top returns bin, screwdriver, pliers, wrench, 2 damp wipe cloths, and a marker
- Conveyor parts tote, orientation nest, and reject bin
- Fitted sheet: no tools; the bed and mattress are the station
- Lever grease gun, correct coupler and hose, 2 color-coded rags, and grease pencil
- Dishwasher draining board and utensil cup
- Mop, mop pad, bucket with wringer, water jug, and caution sign
- Kettle, fill jug, and absorbent cloth
- Pantry stand and collapsible delivery box
- Refrigerator stock tote
- Freezer work stand, work tray, and 2 labeled bins with front rails
- Heat-proof plate, vented cover, food tub, and spoon
- Spray bottle, squeegee, and lint-free cloth
- 3D-printer build plate, debris tray, plastic scraper, and soft brush
- Bookshelf bin, damp cloth, and solid bookend
- Dishwasher unload: three-compartment cutlery drawer insert

- Dryer unload: hamper with hoop carry handle and lint bin
- Pillows and throw: no tools; the bed, pillows, cases, and throw are the station
- Injection mold: flush cutters, parts tray, scrap cup, and reject bin
- Duvet: no tools; the bed, duvet, and cover are the station
- Press brake: angle gauge, gauge mat, finished tray, and reject bin
- Entryway: bowl, shoe rack, and mat are fixed; no tools
- Wardrobe: lidded divided storage box and 2 hangers
- Handheld vacuum: cordless handheld with latching power button, clear dust cup with flap and catch, and its dock; open-top bin

Episode inputs and consumables

- Workbench 5S: 3 strays (mug, stapler, parts box), 2 consumable packs (glove box, wipe pack), and 1 checklist card with an empty four-box row; strays, packs, tools, and cloths reused, plus a scatter of loose bench dust per episode
- Conveyor: 10 tote parts plus 1 belt misfeed part
- Outbound lanes: 6 labelled shoebox-size cases (A1, A2, A3, B1, B2, B3; A3 and B1 weighted with a HEAVY band) and 1 manifest card, all reused
- Fitted sheet: 1 cot-size fitted sheet (reused until its elastic goes slack or a seam opens)
- Greasing: grease for at least 9 strokes and 1 blank lube-card row
- Dishwasher: 12 place-setting items and 1 detergent pod
- Mopping: water to the bucket fill line and 1 clean mop pad
- Sink refill: cold water for the kettle and the dispenser tank, carried by the fill jug
- Pantry: 6 new stock items plus 3 older items for initial placement, and 1 collapsible delivery box (reused until it no longer collapses or stands)
- Refrigerator: 7 new stock items plus 6 older items for initial placement
- Freezer: 6 dated items (3 bags, 3 boxes) and 1 label sheet carrying 2 labels
- Microwave: food for 3 level spoonfuls plus remainder
- Mirror: glass cleaner for at least 6 bursts
- 3D printer: 1 completed print with skirt/purge line and 1 queued job
- Bookshelf: 6 books, 3 non-book objects, and removable shelf dust
- Dishwasher unload: 12 place-setting items (2 plates, 2 bowls, 2 mugs, 6 utensils), reused every episode
- Dryer unload: 6 garments (3 t-shirts, 3 hand towels) reused every episode, and 1 lint sheet per episode
- Pillows and throw: 2 bare pillows, 2 cases, and 1 throw (reused)
- Injection mold: 1 molded part with runner per episode, plus scrap runners for setup test cuts
- Duvet: 1 duvet and 1 buttoned cover (reused until a button or seam fails)

- Press brake: 1 numbered sheet blank per episode (consumed by the bends)
- Entryway: 2 coats, 1 pair of shoes, 1 key set, 1 bag (reused)
- Wardrobe: 4 out-of-season and 4 in-season soft items, 2 hangers (reused)
- Handheld vacuum: 1 scatter of dry crumbs over the rug per episode; rug and vacuum reused, vacuum kept charged

Points to confirm before procurement

1. **Conveyor part count:** the SOP says the tote contains 10 parts and one part is already on the belt. Confirm whether to stage 11 total parts or take the belt misfeed from the original set of 10.
2. **Food and cleaning specifications:** the SOPs define functional properties but do not name an exact food, glass-cleaner formula, grease grade, mop-water additive, detergent-pod brand, or book sizes. Select safe, repeatable standard items and record the chosen specifications.
3. **Reusable versus replaced each episode:** define the replacement frequency for rags, cloths, mop pads, lube cards, detergent pods, food, printed parts, simulated refrigerator and pantry stock, pantry delivery boxes, and freezer label sheets.
4. **Lockout control:** the grease task requires a stopped machine with its isolator at OFF and a lockout tag already fitted. Confirm the authorized lockout procedure and responsible person before collection.